Prodigy Anesthesia Quick Review

Controlled Hypotension

What is 'controlled hypotension'?	The deliberate reduction in systemic blood pressure in order to reduce blood loss *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 958.*
What is traditionally regarded as the lowest mean arterial pressure allowed when performing controlled hypotension?	Although the actual number designated as the lowest allowable pressure is determined on patient history and clinical status, the generally accepted lowest value is 60 mmHg. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 959.*
What criteria should the anesthetist monitor in order to ensure that the patient is perfusing tissues adequately during controlled hypotension?	The urine output, arterial blood gases, and measures of cardiac and cerebral perfusion pressure should be monitored and maintained at normal levels during controlled hypotension. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 959.*
What is the chief anesthetic concern when inducing controlled hypotension in patients with chronic hypertension?	Because patients with chronic hypertension typically require a higher mean arterial pressure in order to maintain adequate perfusion of vital organs. Controlled hypotension may place the patient at risk for ischemia if the pressure is lower than the patient can safely tolerate. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 959.*
What are the advantages and disadvantages of using fenoldapam for controlled hypotension?	Despite the decrease in arterial pressure, fenoldapam markedly increases renal blood flow, urinary flow rate, urinary sodium extraction, and creatinine clearance. It can produce tachycardia and increases in intraocular pressure and should be used with caution in patients with glaucoma. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 261.*
What are the advantages and disadvantages of using sodium nitroprusside to induce controlled hypotension?	Sodium nitroprusside preserves cardiac output well, but can result in reflex tachycardia, rebound hypertension, pulmonary shunting, and carries the risk of cyanide toxicity. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 959.*
What are the advantages and disadvantages of using nitroglycerin to induce controlled hypotension?	Nitroglycerin preserves cardiac perfusion well, but can increase intracranial pressure. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 959.*
What are the advantages and disadvantages of using esmolol to induce controlled hypotension?	It is useful for treating tachycardia, but can potentially result in significant myocardial depression. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 959.*

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Which would be more suitable for producing controlled hypotension, hydralazine or sodium nitroprusside? Why?	Although numerous agents such as calcium channel blockers, volatile agents, and hydralazine are acceptable for producing controlled hypotension, the rapid onset and short duration of action of sodium nitroprusside and nitroglycerin make them most suitable for controlled hypotensive techniques. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 261.*
What are the relative contraindications to performing controlled hypotension?	Relative contraindications to performing controlled hypotension include severe anemia, cardiovascular and cerebrovascular disease, renal or hepatic insufficiency, uncontrolled glaucoma, and hypovolemia. Chronic hypertension is not a relative contraindication to controlled hypotension, but it warrants closer monitoring of blood pressure. Because the alteration of autoregulation of cerebral blood flow in chronically hypertensive individuals, the mean arterial blood pressure should not be lowered more than 20-30% below baseline. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 262.*

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